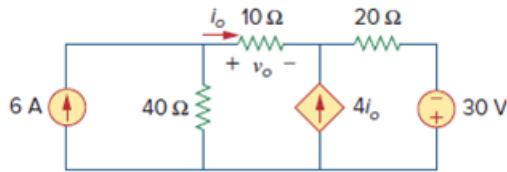


Problem

Use the superposition principle to find i_o and v_o in the circuit of Fig. 4.79.

Figure 4.79:



Step-by-step solution

Step 1 of 9

Refer to the circuit diagram depicted in Figure 4.79 in the textbook.

Circuit has two independent sources, so first consider 6 A current source alone and replace 30 V voltage source with short circuit. Assume that the output current and voltage due to acting of 6 A source be i_o' and v_o' .

Draw the modified circuit diagram.

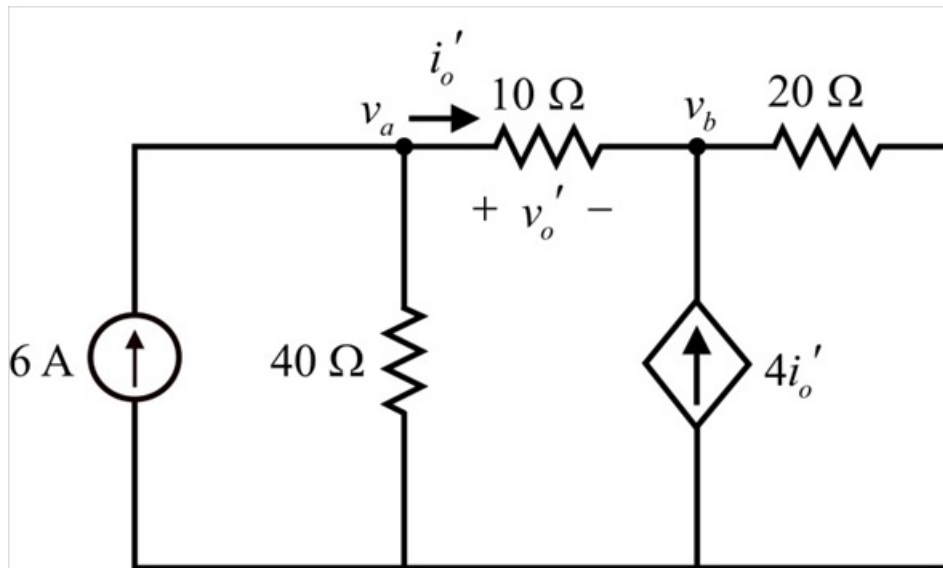


Figure 1

Step 2 of 9

From Figure 1, the value of current i_o' is,

$$i_o' = \frac{v_a - v_b}{10}$$

Apply Kirchhoff's current law at node v_a .

$$-6 + \frac{v_a}{40} + \frac{v_a - v_b}{10} = 0$$

$$\left(\frac{1}{40} + \frac{1}{10}\right)v_a - \frac{1}{10}v_b = 6$$

$$0.125v_a - 0.1v_b = 6$$

$$v_a = \frac{6 + 0.1v_b}{0.125}$$

Step 3 of 9

Apply Kirchhoff's current law at node v_b .

$$\frac{v_b - v_a}{10} - 4i_o' + \frac{v_b}{20} = 0$$

$$\frac{v_b - v_a}{10} - 4\left(\frac{v_a - v_b}{10}\right) + \frac{v_b}{20} = 0 \quad \left(\text{since } i_o' = \frac{v_a - v_b}{10}\right)$$

$$-\frac{1}{10}v_a + \frac{1}{10}v_b - \frac{4}{10}v_a + \frac{4}{10}v_b + \frac{1}{20}v_b = 0$$

$$-0.5v_a + 0.55v_b = 0$$

Substitute $\frac{6 + 0.1v_b}{0.125}$ for v_a .

$$0.55v_b = 0.5v_a$$

$$0.55v_b = 0.5\left(\frac{6 + 0.1v_b}{0.125}\right)$$

$$0.55v_b = 24 + 0.4v_b$$

$$0.15v_b = 24$$

$$v_b = 160 \text{ V}$$

Calculate the value of node voltage, v_a .

$$v_a = \frac{6 + 0.1(160)}{0.125}$$

$$v_a = 176 \text{ V}$$

Step 4 of 9

Determine the value of voltage v_o' .

$$\begin{aligned} v_o' &= v_a - v_b \\ &= 176 - 160 \\ &= 16 \text{ V} \end{aligned}$$

Therefore, the value of output voltage when 6 A source acting is 16 V .

Determine the value of current i_o' .

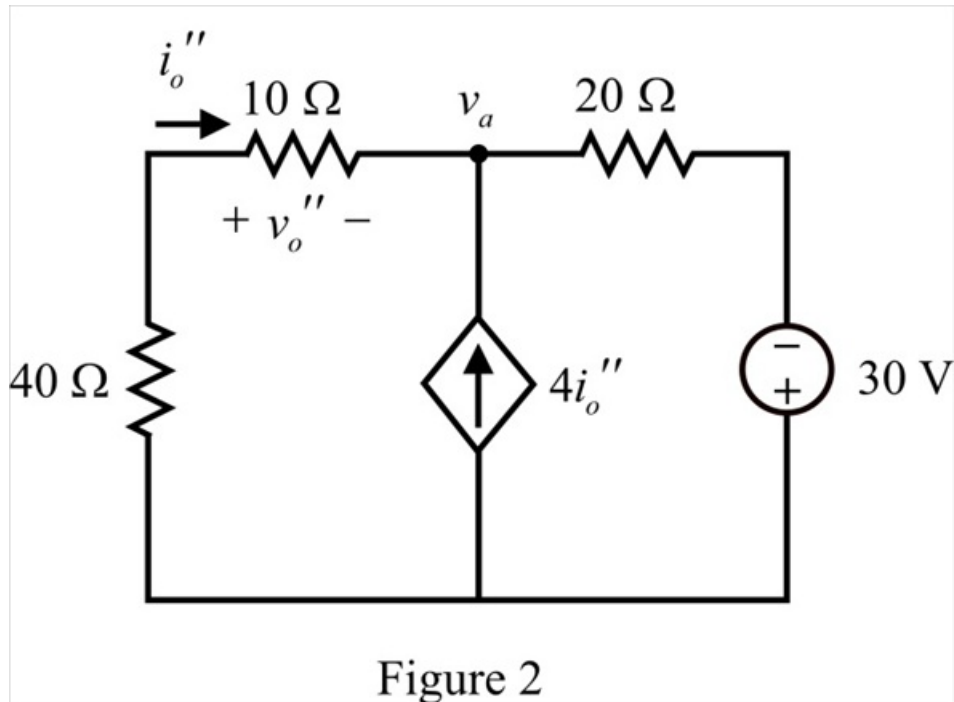
$$\begin{aligned} i_o' &= \frac{v_a - v_b}{10} \\ &= \frac{176 - 160}{10} \\ &= 1.6 \text{ A} \end{aligned}$$

Therefore, the value of output current when 6 A source acting is 1.6 A .

Step 5 of 9

Next consider 30 V voltage source alone and replace 6 A current source with open circuit. Assume that the output current and voltage due to acting of 30 V voltage source be i_o'' and v_o'' .

Draw the modified circuit diagram.



Step 6 of 9

From Figure 2, the value of current i_o'' is,

$$i_o'' = \frac{-v_a}{40 + 10}$$

$$i_o'' = -0.02v_a$$

Apply Kirchhoff's current law at node v_a .

$$\frac{v_a}{50} - 4i_o'' + \frac{v_a + 30}{20} = 0$$

$$\frac{v_a}{50} - 4(-0.02v_a) + \frac{v_a + 30}{20} = 0 \quad \left(\text{since } i_o'' = -0.02v_a \right)$$

$$\frac{v_a}{50} + 0.08v_a + \frac{v_a}{20} = -1.5$$

$$\frac{v_a}{50} + 0.08v_a + \frac{v_a}{20} = -1.5$$

$$0.15v_a = -1.5$$

$$v_a = -10\text{ V}$$

Step 7 of 9

Determine the value of current i_o'' .

$$\begin{aligned} i_o'' &= -0.02v_a \\ &= -0.02(-10) \\ &= 0.2 \text{ A} \end{aligned}$$

Therefore, the value of output current when 30 V source acting is 0.2 A.

Determine the value of voltage v_o'' .

$$\begin{aligned} v_o'' &= 10i_o'' \\ &= (10)(0.2) \\ &= 2 \text{ V} \end{aligned}$$

Therefore, the value of output voltage when 30 V source acting is 2 V.

Step 8 of 9

Determine the total output current when both sources are acting.

$$\begin{aligned} i_o &= i_o' + i_o'' \\ &= 1.6 + 0.2 \\ &= 1.8 \text{ A} \end{aligned}$$

Therefore, the value of output current, i_o in the circuit is 1.8 A.

Step 9 of 9

Determine the total output voltage when both sources are acting.

$$\begin{aligned} v_o &= v_o' + v_o'' \\ &= 16 + 2 \\ &= 18 \text{ V} \end{aligned}$$

Therefore, the value of output voltage, v_o in the circuit is 18 V.